

Did Employer Skill Language Shift After the Generative AI Inflection Point?

Evidence from U.S. Job Posting Language, 2019–2025

Final paper

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EPPS 6323 – Knowledge Mining

Spring 2026

Date: May 8, 2026

Introduction

The rapid spread of generative artificial intelligence after late 2022 created an important turning point in the labor market. The release of ChatGPT in November 2022 made generative AI visible not only to technology firms, but also to employers, workers, universities, and policymakers. Since then, there has been a large public debate about whether AI will replace jobs, reduce wages, or increase productivity. However, one important issue has received less attention: how employers changed the way they describe skills in job postings.

Job postings are useful because they show what employers say they are looking for from workers. They include required skills, duties, qualifications, and sometimes even the kind of worker the employer imagines for a position. If generative AI changed the structure of work, we should be able to see at least some of that change in job description language. This project therefore does not directly study employment loss or wage change. Instead, it studies whether employer skill language shifted after the generative AI inflection point.

The main idea of this paper comes from the task-based theory of technological change. This literature argues that technology does not simply destroy jobs in a mechanical way. Rather, technology changes the tasks inside jobs. Some tasks may be substituted by machines, while other tasks may become more important because they complement technology (Autor, Levy, and Murnane 2003; Acemoglu and Restrepo 2020). In this same logic, generative AI may substitute

for some routine cognitive tasks, while also increasing the value of other skills such as judgment, communication, technical monitoring, and coordination.

However, the direction of change is not obvious. One possible expectation is that high AI-exposed occupations should mention AI and technical skills more after ChatGPT. But another possibility is that these occupations may mention these skills less because employers begin to assume that workers already know how to use AI tools. In that case, AI skill may become an implicit expectation rather than an explicit requirement. Lower-exposure occupations may also start to catch up by adding more AI-related language as generative AI spreads outside traditional technical occupations. For this reason, the effect of generative AI on employer skill language is an empirical question, not something we can assume beforehand.

This study asks the following main research question: Did occupations with higher pre-existing AI exposure experience different post-2022 changes in job posting skill language compared to occupations with lower AI exposure? Three specific questions follow from this. First, did AI-related terminology change after the diffusion of generative AI? Second, did technical skill language change differently across high- and low-AI-exposed occupations? Third, did soft skill and coordination language change after ChatGPT?

To answer these questions, the study uses a large sample of U.S. job postings. The final analytical sample includes 149,912 job postings from 2019–2021 and 2023–2025. The year 2022 is excluded because it is the transition year around the release of ChatGPT, and the final analysis begins the post-period in 2023. The study combines job posting text with the AI Occupational Exposure Index developed by Felten, Raj, and Seamans (2021). This index allows us to compare occupations that were more or less exposed to AI before the generative AI shock.

The empirical strategy uses a continuous difference-in-differences design. The shock is the broad diffusion of generative AI after late 2022. The treatment intensity is occupational AI exposure, measured by AIOE. This means the study does not simply compare treated and untreated jobs. Instead, it examines whether occupations with higher AI exposure changed more after ChatGPT compared to occupations with lower AI exposure. The main outcomes are AI keyword language, technical skill language, and soft skill language in job postings.

This project contributes to the literature in two ways. First, it studies employer expectations directly through job posting language rather than only looking at employment or wages after labor market adjustment has already happened. Second, it connects AI exposure measures with text-based labor market outcomes. This helps us understand whether generative AI is already changing how employers describe skills, even before its full effects appear in wages, productivity, or employment.

Literature Review

Research on technology and labor markets has moved from simple job-loss arguments toward more careful explanations of how technology changes tasks. This is important for our study because job postings are also task and skill documents. They do not only say “we need one worker.” They also describe what that worker should know, do, and bring to the job.

Task-Based Theories of Technological Change

A central contribution in this literature is Autor, Levy, and Murnane (2003). They argue that computerization mainly substitutes for routine tasks, especially tasks that follow clear rules and can be codified. At the same time, computers may complement workers in nonroutine analytical and interpersonal tasks. This means technology does not affect all workers or all skills in the same way. It changes the composition of tasks within occupations.

This task-based view is useful for understanding generative AI. Large language models can write text, summarize information, answer questions, produce code, and support some analytical tasks. These are not only manual tasks. They are cognitive and informational tasks. Therefore, generative AI may affect many white-collar and knowledge-based occupations, not only traditional routine jobs.

Acemoglu and Restrepo (2020) also explain that automation has both displacement and productivity effects. The displacement effect happens when machines or software perform tasks that workers used to do. The productivity effect happens when technology makes production more efficient and creates new demand for other tasks. Both effects can happen at the same time. This

is why the impact of AI may look mixed. Some skills may become less visible in job postings if AI substitutes for them, while other skills may become more visible if workers need to supervise or complement AI tools.

For this study, the key point is that generative AI should not be expected to create only one type of change. It may increase explicit demand for some skills, reduce explicit demand for others, or make some skills so normal that employers stop mentioning them directly.

Social Skills and Technology

Another important part of the literature focuses on social and coordination skills. Deming (2017) shows that social skills became more important in the labor market, especially in jobs that combine technical knowledge with communication and teamwork. His argument is that computers and machines are better at some technical and routine tasks, but they are weaker at social interaction, persuasion, coordination, and relationship management.

This creates an important expectation for generative AI. If AI handles more information-processing tasks, employers may value soft skills more. Workers may need to coordinate teams, communicate with clients, check AI outputs, and make decisions using AI-generated information. In this case, soft skill language may increase after ChatGPT, especially in occupations more exposed to AI.

But there is also another possibility. If generative AI reduces the need to explicitly describe many tasks, job postings may become less detailed about some skills. Employers may assume that workers in high-exposure occupations already understand digital tools and AI-supported workflows. In that case, soft skill language may not increase in high-AIOE jobs. It may even decline relatively, especially if lower-AIOE jobs begin adding more AI and coordination language after 2022. This second possibility is important because it fits the final results of this study more closely.

Measuring Occupational Exposure to Artificial Intelligence

To study AI exposure, this project uses the AI Occupational Exposure Index developed by Felten, Raj, and Seamans (2021). The AIOE index links AI applications to occupational abilities using ONET data. The basic idea is that some occupations depend more heavily on abilities that AI can

support or perform. Occupations such as legal, business and financial operations, and computer and mathematical jobs tend to have higher AI exposure. Manual, service, construction, and transportation-related occupations tend to have lower AI exposure.

A major strength of the AIOE index is that it is measured at the occupation level and is not created from the post-ChatGPT job postings used in this study. This makes it useful as a pre-existing exposure measure. In other words, the exposure measure is not mechanically produced by the outcome we are trying to explain. It gives us a way to ask whether occupations that were already more exposed to AI changed differently after the generative AI shock.

Webb (2020) provides another related way to measure technology exposure. He uses the similarity between patent text and occupational task descriptions to estimate which occupations are more exposed to automation technologies, including AI. His work also suggests that AI exposure is concentrated in high-skill and information-processing occupations. This supports the idea that generative AI may affect white-collar and knowledge work in a different way than earlier waves of automation.

Synthesis and Research Gap

Together, this literature suggests a common argument: technology changes the tasks workers perform, and this should affect the skills employers demand. If generative AI changes workplace tasks, then job postings should show some change in skill language after the diffusion of ChatGPT.

However, existing research has not fully examined whether generative AI has already changed employer skill language at scale. Much of the literature focuses on employment, wages, or occupational exposure. Those outcomes are important, but they may appear after firms and workers have already adjusted. Job postings are different because they can show employer expectations earlier.

This study fills that gap by combining job posting text with occupational AI exposure. It asks whether high-AIOE occupations changed differently from low-AIOE occupations after ChatGPT. It also helps us see whether generative AI creates more explicit demand for AI and technical skills, or whether it reduces explicit skill language in jobs where AI is becoming normalized.

Theory and Hypotheses

Theoretical Framework

This study is based on the task-based theory of technological change. The main logic is simple: technology affects tasks first, and then task changes affect the skills employers ask for. Generative AI is especially important because it can perform many cognitive and language-related tasks. These include writing, summarizing, searching, coding support, drafting, classification, and some forms of analysis.

After ChatGPT became widely known in late 2022, employers may have started to rethink what workers need to do. Some tasks that were previously done directly by workers may now be supported by AI. At the same time, new tasks may appear, such as checking AI output, writing better prompts, using AI tools responsibly, or combining AI results with professional judgment.

But this process should not affect all occupations equally. Occupations with higher AIOE scores were already more connected to AI-related abilities. These occupations should be more likely to experience changes after the generative AI shock. Lower-AIOE occupations may also change, but the change may be slower or different in form.

The key issue is the direction of change. There are three possible mechanisms.

First, there may be a complementarity mechanism. If AI makes workers more productive, employers may demand more technical skills, more AI familiarity, and more soft skills for coordination and oversight. Under this view, high-AIOE jobs should mention AI and related skills more after ChatGPT.

Second, there may be a substitution mechanism. If AI can perform some tasks directly, employers may reduce explicit demand for those tasks in job descriptions. In this view, high-AIOE jobs may show lower technical or soft skill density because AI is expected to handle part of the work.

Third, there may be a normalization mechanism. In high-AIOE jobs, employers may begin to treat AI use as something normal and assumed. If a skill becomes assumed, it may not need to be written in every job posting. This can produce a counterintuitive result: AI-exposed occupations may

mention AI-related skills less, not because AI is unimportant, but because AI has become part of the background expectation.

These mechanisms are not mutually exclusive. Some occupations may show complementarity, while others show substitution or normalization. Therefore, this study treats the direction of change as an empirical question. The important expectation is not that all skill language must increase. Rather, the expectation is that occupations with different levels of AI exposure should show different patterns after ChatGPT.

Hypotheses

H1: AI Keyword Language.

Occupations with higher pre-existing AI exposure will experience differential post-2022 changes in AI-related terminology in job postings compared to lower-exposure occupations.

This hypothesis does not assume that AI keyword language must increase in high-AIOE jobs. One possibility is that employers in high-exposure occupations explicitly mention AI tools more after ChatGPT. However, another possibility is that AI knowledge becomes normalized in those occupations, so employers mention it less directly. Therefore, the direction of the effect is left open.

H2: Technical Skill Language.

Occupations with higher AI exposure will experience differential post-2022 changes in technical skill language compared to lower-exposure occupations.

The expected direction is also ambiguous. If generative AI complements workers, employers may demand more technical skills to use and supervise AI tools. But if AI substitutes for some technical tasks, employers may reduce explicit technical skill requirements. Therefore, a negative effect would be consistent with substitution or normalization, while a positive effect would be more consistent with complementarity.

H3: Soft Skill and Coordination Language.

Occupations with higher AI exposure will experience differential post-2022 changes in soft skill and coordination language compared to lower-exposure occupations.

The complementarity argument suggests that soft skills should become more important because workers may need to communicate, coordinate, and make judgments around AI-generated outputs. However, if AI reduces explicit task requirements or if employers simplify job descriptions in high-exposure occupations, soft skill language may decline relatively. Therefore, this hypothesis also focuses on differential change rather than assuming only an increase.

Together, these hypotheses allow us to study whether generative AI changed employer expectations across three observable dimensions: AI keyword language, technical skill language, and soft skill language. The results are especially important because they can tell us whether AI is mainly creating new explicit skill demands, or whether it is also making some skills implicit, normalized, or substituted in high-exposure occupations.

4. Methodology

(1) Identification Strategy

This study examines whether employer skill language changed after the rapid diffusion of generative artificial intelligence following the release of ChatGPT in late 2022. The core idea behind the identification strategy is that generative AI represented a broad technological shock that affected occupations differently depending on their pre-existing exposure to AI-related tasks. Rather than assuming all occupations experienced the same level of disruption, this study uses differences in occupational exposure to identify whether language in job postings shifted more strongly in occupations that were already more connected to AI capabilities.

The study relies on the AI Occupational Exposure Index (AIOE) developed by Felten, Raj, and Seamans (2021). The AIOE links artificial intelligence applications with occupational abilities using O*NET data and provides a standardized measure of how exposed occupations are to AI technologies. Higher AIOE scores indicate occupations with greater exposure to AI-related tasks,

while lower scores represent occupations that are less directly connected to AI capabilities. Importantly, the AIOE is time-invariant, meaning that the exposure measure itself does not change after the release of ChatGPT. This allows the study to treat generative AI diffusion as an external shock while using pre-existing occupational differences as the source of variation.

The release of ChatGPT in November 2022 serves as the primary inflection point for the analysis. This period marked a visible acceleration in public awareness and organizational adoption of generative AI tools. Employers, workers, and policymakers increasingly began discussing how AI could reshape workplace tasks, productivity, and skill requirements. Because of this major technological shift, the period after late 2022 is treated as the post-treatment period.

The study uses a continuous Difference-in-Differences (DiD) framework. Unlike a traditional DiD design that compares only treated and untreated groups, this approach allows treatment intensity to vary continuously across occupations according to their AIOE score. Occupations with higher AI exposure are expected to experience stronger changes in employer language if generative AI truly altered labor market expectations.

The unit of analysis is occupation by month. The pre-treatment period includes data from 2019 through 2021, while the post-treatment period includes data from 2023 through 2025. The year 2022 was excluded from the final sample because AIOE matching produced relatively low coverage during the transition year surrounding the release of ChatGPT. Excluding 2022 also helps avoid ambiguity and bias regarding whether postings were created before or after the major diffusion of generative AI tools.

The baseline Difference-in-Differences model estimates whether occupations with higher AI exposure experienced larger shifts in skill language after the generative AI shock. The empirical specification includes occupation fixed effects and year fixed effects. Occupation fixed effects control for time-invariant differences across occupations, while year fixed effects account for broader macroeconomic or labor market trends affecting all occupations over time.

The main estimating equation is:

$$Y_{it} = \alpha + \beta(Post_t \times AIOE_i) + \gamma_i + \delta_t + \epsilon_{it}$$

where Y_{it} represents the outcome variable, including AI keyword density, technical skill density, or soft skill density. The variable ($Post_t$) captures the post-2022 period, while ($AIOE_i$) measures occupational exposure to AI. The interaction term between the post period and AIOE captures the differential effect of generative AI diffusion on occupations with varying levels of AI exposure.

The coefficient of interest is (β). A statistically significant coefficient would suggest that occupations with higher exposure to AI experienced different patterns of employer skill language after the release of ChatGPT. This framework allows the study to move beyond simple descriptive trends and evaluate whether changes were systematically associated with AI exposure.

To further evaluate the validity of the research design, the study also employs an event-study specification. The event-study analysis tests for differential pre-trends across occupations before the release of ChatGPT. If occupations with high and low AI exposure followed similar trends before 2022, then the parallel trends assumption required for Difference-in-Differences analysis becomes more plausible.

The event-study results show that pre-period coefficients between 2019 and 2021 remain close to zero and statistically insignificant. This suggests that occupations with different levels of AI exposure were moving similarly before the technological shock. After 2022, however, the coefficients begin to diverge, indicating that the shift in employer language emerged after the diffusion of generative AI technologies rather than before.

Overall, the identification strategy combines a major technological shock with pre-existing occupational exposure differences to evaluate how employer language changed during the rise of generative AI. By integrating the AIOE measure with a continuous Difference-in-Differences design, the study provides a structured way to analyze whether labor market expectations shifted systematically across occupations.

(2) Knowledge Mining and Text Analysis

The study uses large-scale online job posting data collected from multiple employment platforms between 2019 and 2026. The original raw dataset contained approximately 328,500 job postings gathered from platforms including Indeed, LinkedIn, Upwork, Google Jobs, Greenhouse, Lever, Jobicy, The Muse, Open Science Framework (OSF), and USA Jobs. These platforms were selected because they provide broad coverage across industries, occupations, and skill levels within the U.S. labor market.

After collection, the raw postings were matched with occupational categories using the AI Occupational Exposure Index (AIOE). The matching process achieved approximately a 60 to 70 percent match rate depending on the year and platform. Following data cleaning, occupation matching, and filtering procedures, the final analytical sample consisted of 149,912 job postings across 22 occupational categories. We try to use Indeed data sources as main source but we need to use others while Indeed isn't available.

The final sample included 26,094 postings from 2019, 29,967 from 2020, 29,939 from 2021, 13,443 from 2023, 14,362 from 2024, and 36,107 from 2025. These postings formed the basis for the text mining and econometric analyses conducted in the study.

The knowledge mining component of this project transformed unstructured job posting text into structured outcome variables suitable for quantitative analysis. The original job postings contained information such as job titles, job descriptions, posting years, platform sources, occupational classifications, SOC codes, and AIOE exposure values. Because the central research question concerns changes in employer skill language, the text mining process focused on extracting measurable indicators from the language used in job titles and job descriptions.

The first step was to prepare the text for analysis. For each job posting, the job title and job description were combined into a single text field. This was important because job titles alone are often too short to capture the full range of skills and responsibilities associated with a position, while job descriptions contain more detailed information about employer expectations. Combining both fields allowed the analysis to capture both the formal role label and the broader language of required tasks, qualifications, and skills.

The second step involved text cleaning and standardization. The combined text was converted to lowercase, punctuation and unnecessary symbols were removed, and spacing was standardized. These steps reduced formatting inconsistencies across job postings collected from different platforms and years. This was necessary because the dataset came from multiple job-posting sources, and raw text often contained inconsistent formatting, symbols, and other irregularities.

The third step was tokenization. Tokenization breaks text into individual words or terms that can be counted and analyzed systematically. After tokenization, common stop words and irrelevant terms were removed so that the analysis focused on substantively meaningful job-posting language. Additional cleaning was conducted to remove platform artifacts and formatting-related terms that appeared in some years, such as HTML-related words.

After preprocessing, the project used a dictionary-based text mining approach to create outcome variables. Three keyword dictionaries were manually constructed for this study. The first dictionary captured AI-related language, including terms such as “artificial intelligence,” “AI,” “machine learning,” “deep learning,” “natural language processing,” “NLP,” “large language model,” “LLM,” “generative AI,” “ChatGPT,” “automation,” “algorithm,” “predictive,” and “modeling.” The second dictionary captured technical-skill language, including terms related to data, analytics, statistics, Python, SQL, databases, cloud systems, software, programming, coding, cybersecurity, dashboards, visualization, and technology. The third dictionary captured soft-skill and coordination language, including terms such as communication, collaboration, teamwork, leadership, management, coordination, problem solving, critical thinking, adaptability, interpersonal skills, customer service, support, presentation, and writing.

These dictionaries were not designed to be perfectly mutually exclusive. Some concepts, such as “machine learning,” “algorithm,” or “modeling,” may reasonably belong to both AI-related and technical-skill language. The purpose of the dictionaries was not to create rigid conceptual boundaries, but to capture three different dimensions of employer language: AI visibility, technical skill emphasis, and soft/coordination skill emphasis.

For each job posting, the number of matched terms from each dictionary was counted. To account for differences in job description length, the keyword counts were divided by the total number of words in the combined job title and description text. This produced three density-based outcome

variables: AI keyword density, technical skill density, and soft skill density. Density measures are preferable to raw counts because longer job descriptions naturally contain more words and therefore have more opportunities to include dictionary terms.

These text-based outcome variables were then merged with the existing occupational and exposure variables, including year, source platform, SOC code, AIOE category, and continuous AIOE score. The final modeling-ready file therefore retained job-level observations while adding structured measures of employer skill language. This allowed the team to connect knowledge mining outputs with the Difference-in-Differences and event-study framework used in the empirical analysis.

Descriptive figures from the text-mining pipeline are reported in Appendix Figures A1–A3. Appendix Figure A1 shows the most frequent cleaned terms by year, Appendix Figure A2 shows the yearly share of postings mentioning at least one AI-related keyword, and Appendix Figure A3 compares yearly technical-skill and soft-skill keyword density.

Overall, the knowledge mining process converted raw job posting language into measurable text outcomes. This step was central to the project because the dependent variables used in the empirical analysis were not directly available in the raw data. They had to be constructed from job posting text.

(3) Difference-in-Differences Methodology

The Difference-in-Differences methodology is central to this research because it allows the study to estimate how employer skill language changed after the rise of generative AI while accounting for broader labor market trends. The purpose of the DiD framework is to isolate the relationship between AI exposure and changes in job posting language following the release of ChatGPT.

Traditional descriptive analysis alone cannot determine whether observed changes are actually related to generative AI. For example, technical skill language or AI-related terminology may increase over time because of broader digital transformation trends rather than because of ChatGPT specifically. The DiD approach helps address this issue by comparing occupations with different levels of AI exposure before and after the generative AI inflection point.

In this study, the treatment is not binary. Instead, treatment intensity varies continuously across occupations through the AIOE index. Occupations such as legal professions, business and financial operations, and computer and mathematical occupations generally have higher AIOE scores because many of their tasks overlap with AI capabilities. In contrast, occupations such as construction, transportation, and food preparation tend to have lower AIOE scores because their tasks are less directly exposed to generative AI.

The post-treatment period begins in 2023, immediately following the widespread public diffusion of ChatGPT and related generative AI technologies. By interacting the post period with AIOE scores, the DiD model estimates whether occupations with higher AI exposure experienced larger changes in employer language compared to occupations with lower exposure.

Three main outcome variables are analyzed in the model. The first outcome variable is AI keyword density, which measures the frequency of AI-related terminology within job postings. This includes keywords such as “artificial intelligence,” “machine learning,” “deep learning,” “ChatGPT,” “automation,” and “algorithm.” The second outcome variable is technical skill density, which captures references to technical competencies such as Python, SQL, analytics, cloud computing, programming, and cybersecurity. The third outcome variable is soft skill density, which includes communication, teamwork, leadership, management, adaptability, and problem-solving language.

The dictionaries used in the study are intentionally not mutually exclusive because many job-related concepts overlap. For example, terms such as “modeling” or “machine learning” may reasonably appear in both AI-related and technical-skill categories. The goal of the dictionaries is not to create perfectly separate categories but rather to capture different dimensions of employer language.

The DiD regression results reveal several important findings. First, occupations with higher AI exposure experienced relatively lower soft skill density after the diffusion of generative AI. Second, highly exposed occupations also showed relatively lower technical skill density. Finally, occupations with higher AI exposure became relatively less likely to explicitly mention AI-related keywords after the release of ChatGPT.

At first glance, these findings may appear counterintuitive because one might expect highly AI-exposed occupations to increase their use of AI language after ChatGPT. However, the results may instead reflect a normalization effect. In highly exposed occupations, employers may increasingly assume that workers already possess familiarity with AI tools, reducing the need to explicitly mention these skills in job postings. In contrast, lower-exposure occupations may have begun incorporating more AI-related language as generative AI became more widespread across industries.

The findings also suggest the possibility of skill convergence across occupations. Before the generative AI period, highly exposed occupations may have emphasized technical and AI-related skills much more strongly than lower-exposure occupations. After ChatGPT, however, lower-exposure occupations appear to move closer toward higher-exposure occupations in terms of AI language and skill expectations.

An important strength of the DiD framework is the inclusion of fixed effects. Occupation fixed effects account for stable differences across occupational groups, while year fixed effects capture overall labor market changes such as economic recovery, remote work transitions, or macroeconomic uncertainty. By controlling for these factors, the model focuses more directly on differential changes associated with AI exposure.

The study also clusters standard errors at the occupation level. This adjustment is important because observations within occupations may be correlated over time. Clustering helps produce more reliable standard error estimates and improves the validity of statistical inference.

Despite these strengths, the methodology still has limitations. Because the analysis uses occupation-level exposure rather than direct firm-level AI adoption data, the results cannot fully capture how individual organizations implemented generative AI tools internally. Additionally, the exclusion of 2022 limits the ability to observe immediate short-term reactions during the exact release period of ChatGPT.

Nevertheless, the Difference-in-Differences framework provides a strong empirical approach for studying how employer language shifted during the rise of generative AI. By combining

occupational AI exposure with longitudinal job posting data, the study offers evidence that labor market expectations evolved differently across occupations after the generative AI shock.

4. Results and Discussion

4.1 Knowledge Mining

The descriptive text mining results show that employer language changed meaningfully across the study period. The first descriptive pattern concerns AI-related terminology. The share of job postings mentioning at least one AI-related keyword declined slightly between 2019 and 2021, but increased after the generative AI inflection point. AI keyword mentions rose in 2023 and reached their highest level in 2024 before declining in 2025. Although the 2025 decline suggests that the trend was not a simple permanent increase, AI-related language remained more visible than in the immediate pre-ChatGPT period.

This pattern is important because it shows that generative AI-related terminology became more common in job postings after the public diffusion of ChatGPT. However, this descriptive trend should not be interpreted as causal evidence by itself. It shows that AI language became more visible over time, but the DiD and event-study models are needed to evaluate whether these changes differed systematically by occupational AI exposure.

Appendix Figure A2 shows that AI-related keyword mentions declined slightly from 2019 to 2021, increased in 2023 and 2024, and then declined in 2025 while remaining above the 2021 level. This pattern provides descriptive motivation for the formal DiD analysis, but it should not be interpreted as causal evidence by itself.

The second descriptive pattern concerns technical-skill and soft-skill language. Soft-skill density was consistently higher than technical-skill density across all years. This is expected because many job descriptions include broad workplace terms related to communication, teamwork, management, support, and customer service. Technical-skill density was lower overall, reflecting more specific language related to data, software, programming, cybersecurity, and analytics.

Both technical-skill density and soft-skill density increased after 2021 and peaked around 2023. After 2023, both measures declined somewhat, with the decline more visible for technical-skill density. This suggests that job-posting language shifted not only in terms of explicit AI-related terminology, but also in broader descriptions of skill requirements. The rise and fall of these measures also supports the need for a more formal empirical design, because descriptive trends alone cannot distinguish between general labor market changes, platform differences, and AI-related restructuring.

Appendix Figure A3 shows that soft-skill keyword density remained consistently higher than technical-skill keyword density across all years. Both measures increased after 2021 and peaked around 2023 before declining somewhat in later years. This supports the idea that employer skill language shifted over time, even before accounting for differences in occupational AI exposure.

The keyword-density variables created through the text mining process also serve as the main outcomes for the empirical models. AI keyword density captures the visibility of AI-related terms in job postings. Technical skill density captures the relative presence of technical and computing-related skill language. Soft skill density captures communication, coordination, leadership, and workplace-interaction language. Together, these measures allow the project to examine whether the diffusion of generative AI changed the way employers describe skills across occupations.

The descriptive results align with the broader empirical finding that skill-language changes were not uniform across occupations. As reported in the DiD results, higher-AIOE occupations showed relatively lower post-ChatGPT changes in soft skill density, technical skill density, and AI keyword mentions compared with lower-AIOE occupations. This does not necessarily mean that AI became less important in high-exposure occupations. Instead, it may suggest a convergence or normalization pattern: lower-AIOE occupations began adopting more AI and skill-related language, while high-AIOE occupations may have treated some AI-related skills as implicit expectations.

4.2 Descriptive Statistics

Table 1 presents summary statistics for the full sample of 149,912 job postings starting from 2019 to 2021 (pre-ChatGPT) and 2023 to 2025 (post-ChatGPT). For descriptive purposes, occupations

are classified as high or low AI-exposed based on the median AIOE score (median = 0.637), accounting for 73,011 high AIOE and 76,901 low AIOE job postings. Note that in the regression analysis, AIOE is treated as a continuous variable. The AIOE index ranges from -1.34 to 1.31 with a mean of 0.267 ($SD = 0.788$), reflecting meaningful variation in AI exposure across occupations. High AIOE jobs exhibit systematically higher mean soft skill density (0.0146 vs. 0.0118), technical skill density (0.0071 vs. 0.0036), and AI keyword mentions (20.0% vs. 13.7%) compared to low AIOE jobs, confirming that occupational AI exposure is associated with more skill-intensive job postings in the pre-period. Comparing pre- and post-ChatGPT periods, mean soft skill density increased from 0.0123 to 0.0143 , technical skill density rose modestly from 0.0051 to 0.0055 , and the share of postings mentioning any AI keyword increased substantially from 13.6% to 21.0% , suggesting broad shifts in job posting language following ChatGPT's release.

Table 1: Summary Statistics

Group	N	Mean AIOE	SD AIOE	Mean Soft Skill	SD Soft Skill	Mean Tech Skill	SD Tech Skill	Mean AI Keyword	SD AI Keyword
Panel A: Full Sample									
Full Sample	149,912	0.267	0.788	0.0132	0.0094	0.0053	0.0087	0.168	0.374
Panel B: By AIOE Group									
High AIOE	73,011	0.919	0.194	0.0146	0.0097	0.0071	0.0100	0.200	0.400
Low AIOE	76,901	-0.353	0.623	0.0118	0.0090	0.0036	0.0068	0.137	0.344
Panel C: By Period									

Post-

ChatGPT 63,912 0.229 0.799 0.0143 0.0091 0.0055 0.0085 0.210 0.407
(2023–2025)

Pre-ChatGPT 86,000 0.295 0.779 0.0123 0.0096 0.0051 0.0089 0.136 0.343
(2019–2021)

4.3 Descriptive Trends

Figure 1 plots mean outcome variables by year separately for high and low AIOE occupations, where high AIOE is defined as above the median AIOE score of 0.637, with a vertical dashed line marking ChatGPT’s release in November 2022. Note that each panel has its own y-axis scale reflecting the different units of measurement across outcomes – soft skill density is log-transformed, technical skill density is expressed as a raw winsorized proportion, and AI keyword mentions are expressed as a percentage of postings.

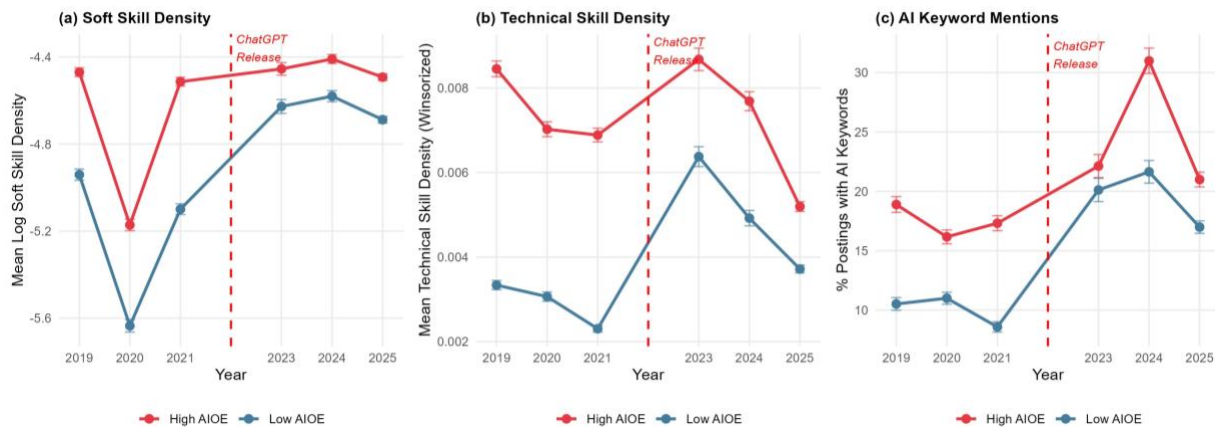


Figure 1: Skill Requirements by AI Exposure Over Time.

Notes: Red line = High AIOE occupations ($AIOE > 0.637$). Blue line = Low AIOE occupations ($AIOE \leq 0.637$). Dashed vertical line marks ChatGPT release (November 2022). Error bars = 95% CI.

Several patterns are immediately apparent. First, as shown in panel (a), high AIOE jobs consistently maintain higher soft skill density throughout the pre-period, with both groups

exhibiting a sharp decline in 2020 — likely reflecting COVID-19 labor market disruptions — followed by recovery in 2021. After ChatGPT’s release, low AIOE jobs show a pronounced upward convergence toward high AIOE jobs, substantially narrowing the pre-existing gap. Second, panel (b) shows that for technical skill density, both groups exhibit stable pre-period trends before a notable spike in 2023 followed by a gradual decline through 2025. The gap between high and low AIOE jobs, while persistent, narrows in the post-period. Third, panel (c) shows that for AI keyword mentions, high AIOE jobs consistently show higher rates throughout the sample period. Following ChatGPT’s release, both groups exhibit sharp increases in AI keyword mentions, with high AIOE jobs peaking at 31% in 2024 before declining in 2025. Low AIOE jobs show a stronger relative increase in the post-period, again suggesting convergence. Together these descriptive patterns provide initial evidence that ChatGPT’s release differentially affected skill requirements across occupations with different levels of AI exposure.

4.4 Parallel Trends Assumption

To validate the identifying assumption under our DiD estimator, we estimate event study models interacting year fixed effects with the continuous AIOE score, using 2021 as the base year. Event study helps us to capture the differential relationship between AIOE and each outcome in any relative to the base year 2021. Under the parallel trends assumption, pre-period coefficients should be statistically indistinguishable from zero.

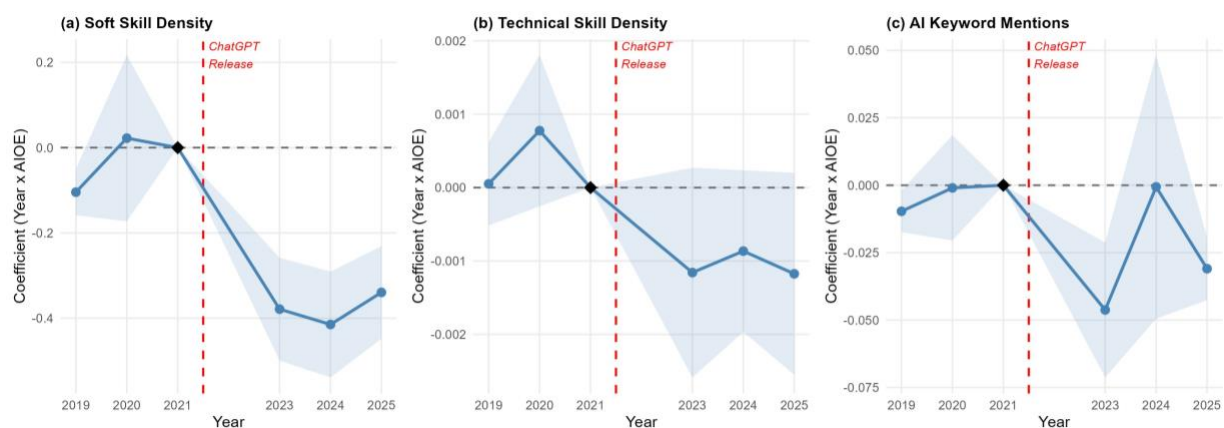


Figure 2: Event Study — Parallel Trends Validation.

Notes: Coefficients from regression of outcome on Year $\times$ AIOE interactions with occupation and year fixed effects. Base year = 2021 (black diamond, normalized to zero). Shaded area = 95% CI.

Clustered SEs at occupation level. Pre-period coefficients close to zero support the parallel trends assumption.

Figure 2 presents the event study plots for all three outcomes. In all three panels, pre-period coefficients for 2019 and 2020 are close to zero and the shaded confidence intervals cross the zero line, indicating that high and low AIOE occupations followed similar trends before ChatGPT's release. This supports the parallel trends assumption across all three outcomes. We also note that year 2022 – the exact year of ChatGPT's release – is absent from our dataset, preventing direct observation of the immediate shock period. Our post-period estimates therefore begin from 2023 onward, capturing the medium-term labor market response. In all three models, post-period coefficients drop consistently below zero and are of substantially larger magnitude than the pre-period coefficients, indicating a clear differential effect following ChatGPT's release.

4.5 Main DiD Results

Table 2 presents the main DiD estimates. The key coefficient of interest is $\text{AIOE} \times \text{Post}$, which captures how skill requirements changed differentially for higher AI-exposed occupations following ChatGPT's release. All models include occupation and year fixed effects, with standard errors clustered at the occupation level.

Table 2: Main DiD Results

	(1) Soft Skill (Log)	(2) Technical Skill (Win.)	(3) AI Keyword (Binary)
AIOE $\times$ Post (DiD)	-0.3378***	-0.0014*	-0.0236**
	(0.0407)	(0.0006)	(0.0071)
Fixed Effects			
Occupation FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Clustered SEs	Occupation	Occupation	Occupation
Observations	149,912	149,912	149,912
R ²	0.096	0.166	0.045
Within R ²	0.008	0.004	0.001

*Notes: Standard errors clustered at the occupation level in parentheses. All models include occupation and year fixed effects. The main effects of AIOE and Post are absorbed by the fixed effects. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.*

The results reveal a consistent and statistically significant negative effect of AI exposure on skill requirements across all three outcomes following ChatGPT's release. For soft skill density, the DiD coefficient is -0.338 ($p < 0.01$), indicating that higher AI-exposed occupations exhibit significantly lower soft skill density in job postings after ChatGPT compared to lower AI-exposed occupations. For technical skill density the coefficient is -0.0014 ($p < 0.05$). Numerically, relative to the pre-period mean technical skill density of 0.0051, this represents a 26.6 percent decline. For AI keyword mentions, the DiD coefficient is -0.024 ($p < 0.01$) in a linear probability model, implying that higher AI-exposed jobs are 2.36 percentage points less likely to mention any AI keyword per unit increase in AIOE following ChatGPT's release.

4.6 Discussion

The results present a coherent and somewhat counterintuitive picture of how ChatGPT reshaped skill requirements in job postings. Across all three outcome dimensions — soft skills, technical skills, and AI keyword mentions — higher AI-exposed occupations show relatively lower skill requirements following ChatGPT's release compared to lower AI-exposed occupations. This pattern is consistent with an AI substitution mechanism, whereby the availability of generative AI tools reduces the need for employers to explicitly demand certain skills, as AI is expected to handle those tasks directly. The finding that higher AI-exposed jobs become less likely to mention AI keywords after ChatGPT is particularly notable – rather than signaling increased demand for AI

literacy, it suggests that AI competence became an implicit assumption in those occupations, no longer requiring explicit mention in job postings.

The convergence pattern observed across groups is also very worth to mention about. Lower AI-exposed occupations exhibit stronger relative increases in soft skill density and AI keyword mentions in the post-period, narrowing the gap with their high AI-exposed counterparts. This convergence suggests that ChatGPT's release diffused AI-related skill demands more broadly – occupations previously less exposed to AI began incorporating AI-related language and soft skill requirements at a faster rate. Taken together, our findings suggest that generative AI did not simply increase skill demand in AI-exposed jobs. Instead, it appears to have reduced explicit skill requirements in higher AI-exposed occupations while lower AI-exposed occupations gradually caught up. This points to a broader restructuring of the labor market – where AI substitutes for skills at the top while spreading skill demand more evenly across occupations.

5. Policy Implications

The findings from this study have several important policy implications for education systems, workforce development programs, employers, and policymakers. One of the main conclusions is that generative AI may be reshaping employer expectations across occupations, even in jobs that were traditionally considered less exposed to artificial intelligence.

The evidence suggests that lower AI-exposed occupations increasingly incorporated AI-related and skill-related language after the release of ChatGPT. This indicates that generative AI is not only affecting highly technical occupations but is spreading across a broader range of industries and work environments. As a result, workforce development programs may need to expand AI-related training beyond traditional STEM fields.

Educational institutions may also need to reconsider how technical and soft skills are integrated into curricula. Rather than teaching AI as a narrow specialized skill, universities and training programs may need to focus on broader digital literacy, adaptability, communication, and human-AI collaboration. Workers increasingly may be expected to interact with AI systems regardless of occupation.

The findings also raise questions about how employers define and communicate skill requirements. If highly AI-exposed occupations are reducing explicit mentions of technical and AI-related skills, this may indicate that employers increasingly assume baseline familiarity with AI tools. Such normalization could create hidden expectations within the labor market, particularly for workers who lack access to AI training or digital resources.

Policymakers should also consider how generative AI may widen inequalities between workers who can adapt to AI-enhanced environments and those who cannot. Access to digital infrastructure, AI literacy programs, and reskilling opportunities may become increasingly important for reducing labor market inequality.

In addition, public workforce programs may need to focus not only on technical training but also on transferable skills such as adaptability, communication, and critical thinking. As AI automates routine tasks, human-centered skills may become increasingly valuable in coordinating, interpreting, and managing AI-supported work processes.

Overall, the results suggest that generative AI is contributing to broader restructuring within labor markets. Policymakers and educators therefore need to prepare workers for evolving skill expectations rather than focusing only on job displacement narratives.

6. Limitations

Although the study provides important insights into changing employer language during the rise of generative AI, several limitations should be acknowledged.

First, the study relies on job posting text rather than direct measures of actual workplace behavior or hiring outcomes. Job advertisements reflect employer communication and expectations, but they may not perfectly represent how work tasks are ultimately performed within organizations.

Second, the analysis uses occupational exposure categories based on the AIOE index rather than direct measures of firm-level AI adoption. Some firms may have adopted generative AI extensively, while others within the same occupation category may have used little or no AI technology.

Third, the exclusion of 2022 limits the ability to observe the immediate transition period surrounding the release of ChatGPT. Because 2022 represents the exact timing of the shock, future studies with more complete monthly data may provide more precise estimates of short-term adjustment effects.

The knowledge mining component transformed unstructured job posting text into structured outcome variables for the empirical analysis. For each posting, the job title and job description were combined into one text field. The text was then cleaned by converting it to lowercase, removing punctuation and unnecessary symbols, standardizing spacing, and tokenizing it into analyzable terms. After removing common stop words and formatting artifacts, the project applied three manually constructed dictionaries: an AI keyword dictionary, a technical-skill dictionary, and a soft-skill dictionary.

For each job posting, the analysis counted the number of matched terms from each dictionary. These counts were then divided by the total number of words in the combined job title and description text to create density measures. The three final text-based outcome variables were AI keyword density, technical skill density, and soft skill density. These variables were then combined with year, source platform, SOC code, AIOE category, and continuous AIOE exposure so that they could be used in the Difference-in-Differences and event-study models.

Descriptively, AI-related keyword mentions declined slightly from 2019 to 2021, increased clearly in 2023 and 2024, and declined again in 2025 while remaining above the 2021 level. Soft-skill density was consistently higher than technical-skill density across all years, and both measures rose after 2021 before declining somewhat after 2023. These patterns show that employer language changed over time, but the descriptive trends alone are not causal evidence. The DiD and event-study models are needed to examine whether these changes differed systematically by occupational AI exposure.

The study is also limited by occupational aggregation. While the research includes 22 occupational categories, there may still be important heterogeneity within occupations that cannot be fully captured at the aggregate level.

Finally, the analysis focuses primarily on U.S. job postings. Labor market responses to generative AI may differ across countries depending on economic conditions, educational systems, regulatory environments, and technology adoption patterns.

7. Future Research Potential

Future research could expand this study in several important ways. One potential direction is the use of higher-frequency monthly or quarterly data to better capture the timing of employer adjustments immediately following major AI developments.

Researchers could also examine heterogeneity within occupational groups to determine whether certain industries, firms, or worker categories experienced stronger changes in employer language. Firm-level AI adoption measures could provide a more detailed understanding of how organizations integrate generative AI into workplace practices.

Another important area for future research is vocabulary analysis beyond predefined keyword dictionaries. More advanced natural language processing techniques, including transformer-based language models, could help identify subtle shifts in employer communication patterns.

Future studies may also examine how generative AI affects wage structures, hiring outcomes, occupational mobility, and worker productivity. Combining job posting data with labor market outcomes could provide a more complete picture of how AI influences employment dynamics.

Cross-national comparisons would also be valuable. Different countries may experience varying labor market responses depending on technological infrastructure, labor regulations, and educational systems.

Finally, future research could investigate how workers themselves respond to changing employer expectations. Surveys, interviews, and worker-level data may help explain whether employees perceive increasing pressure to adopt AI-related skills.

8. AI Usage Disclosure

Artificial intelligence tools were used throughout the research process as supportive research accelerators. AI systems including Claude Sonnet, Gemini, and ChatGPT assisted with brainstorming, coding support, literature summarization, language polishing, and presentation organization.

AI tools were used to brainstorm and identify relevant datasets, research directions, prior studies, and debugging within the text mining pipeline. These tools also assisted with improving transitions, grammar, and writing clarity during the drafting process.

However, all major analytical decisions, interpretations, empirical conclusions, and final arguments were determined by the authors themselves. AI tools were not used to independently generate findings or substitute for scholarly judgment.

The research team-maintained responsibility for verifying all statistical outputs, interpreting regression results, constructing methodological decisions, and evaluating theoretical implications. AI tools functioned primarily as productivity and organizational aids rather than autonomous research agents.

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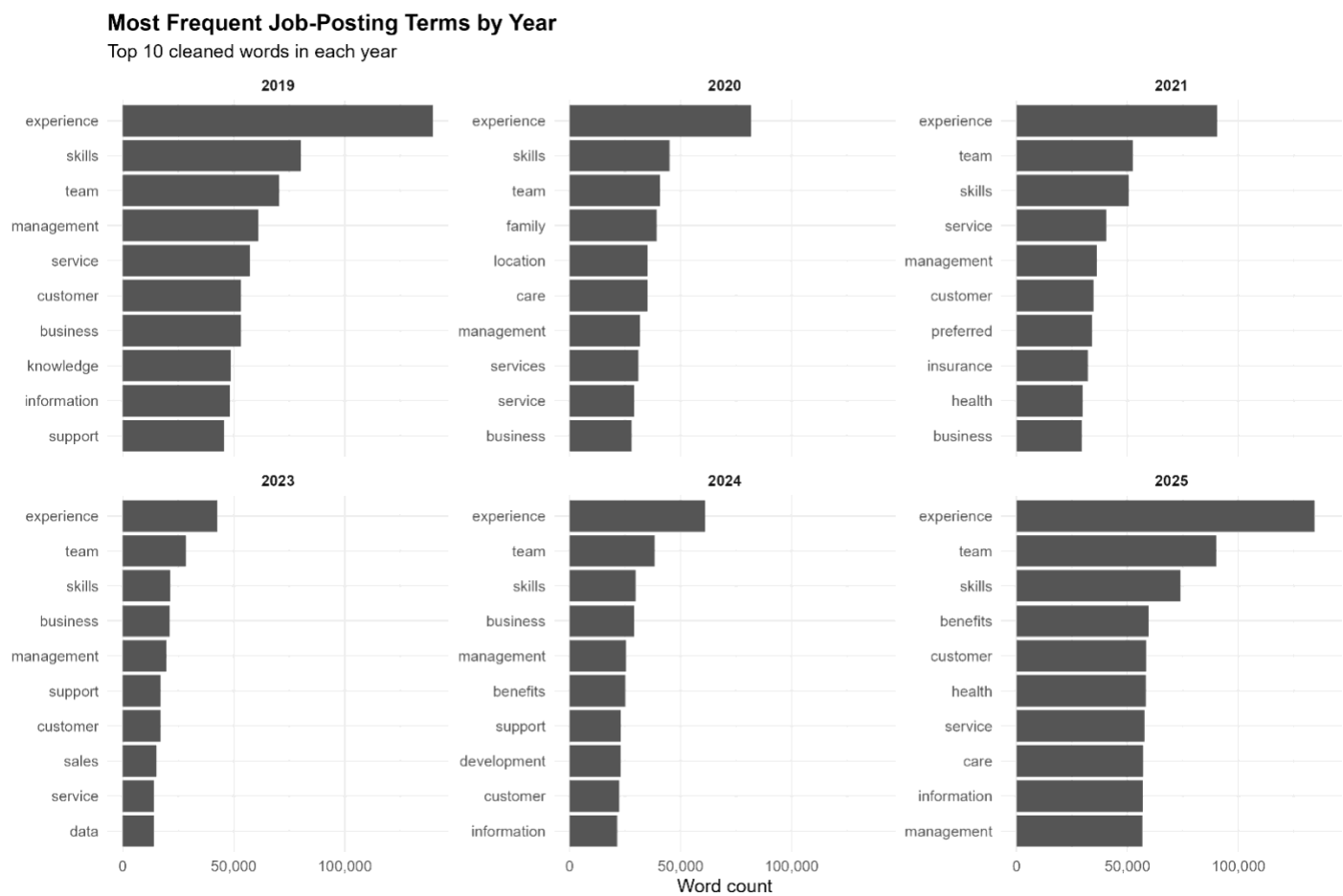
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Appendix Figure A: Text Mining Figures

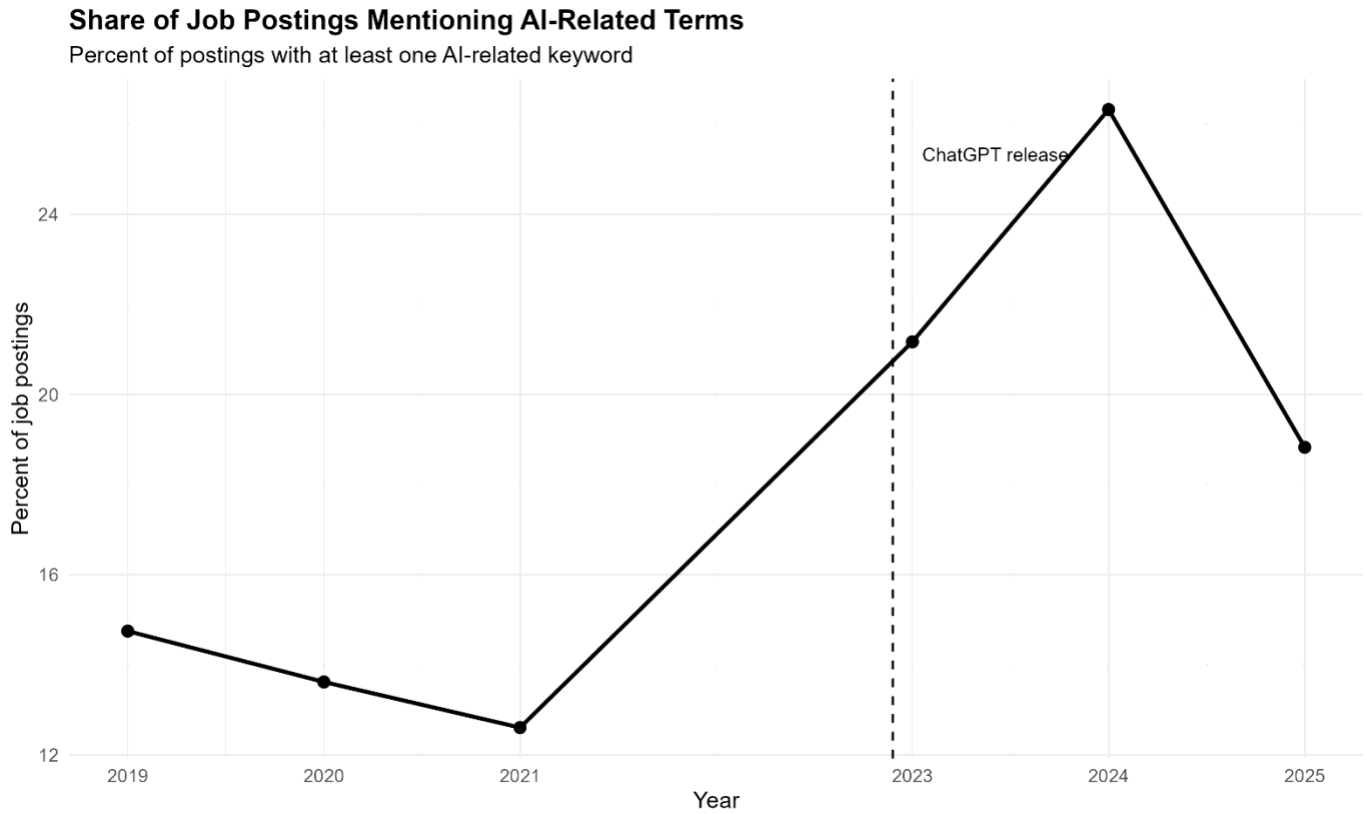
These appendix figures are descriptive and are intended to show the text-mining patterns underlying the constructed outcome variables. The formal test of differential change by AI exposure is provided by the event-study and DiD models.

Figure A1. Most Frequent Job-Posting Terms by Year



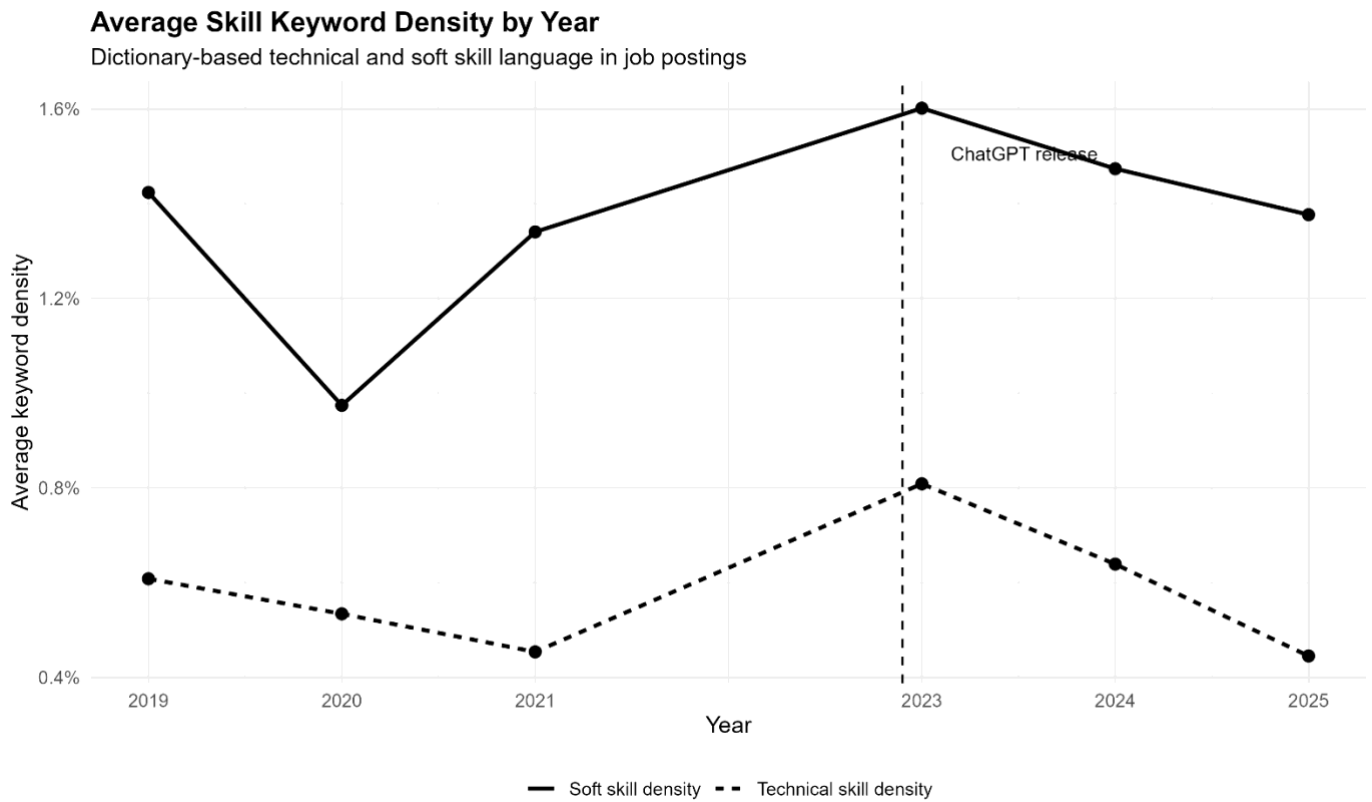
Notes: Figure reports the top 10 cleaned terms for each year after text preprocessing, tokenization, stop-word removal, and removal of formatting artifacts.

Figure A2. Share of Job Postings Mentioning AI-Related Terms by Year



Notes: Figure reports the percentage of postings containing at least one AI-related dictionary term. The dashed vertical line marks the ChatGPT release period.

Figure A3. Average Skill Keyword Density by Year



Notes: Figure reports average dictionary-based technical-skill and soft-skill keyword density. Density is calculated as keyword matches divided by total words in the combined job title and description text.